

 AWS HANDS-ON SERIES · DAY 3

Terraform Outputs Explained

Project 3.1 — Phase 3 · Exporting Values from Infrastructure

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 Terraform v1.7.5

 AWS S3 + SSM

 Swipe to see all 4 output types

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WHAT ARE OUTPUTS?

Export Values After **Apply**

SIMPLE STRING

```
bucket_name = "outputs-demo-e63a1b07"
```

SENSITIVE

```
db_password = <sensitive> — hidden always
```

OBJECT

```
bucket_info = { name, arn, region }
```

COMPUTED

Built from resource values — URLs, commands

-RAW FLAG

No quotes — perfect for shell scripts

-JSON FLAG

Only way to reveal sensitive output

4 OUTPUT TYPES

In This **main.tf**

● Simple String

```
output "bucket_name" { value = aws_s3_bucket.app.bucket }
```

Shows: outputs-demo-e63a1b07

● Sensitive

```
output "db_password" { value = ... sensitive = true }
```

Shows: <sensitive> — hidden in ALL terminal output

● Object

```
output "bucket_info" { value = { name, arn, region } }
```

Shows: grouped map with 3 fields

● Computed URL

```
output "bucket_console_url" { value = "https://.../${bucket}" }
```

Shows: direct AWS Console link

TERRAFORM APPLY

All 4 Outputs **Live**

```
$ terraform apply - Outputs section

bucket_console_url = "https://s3.console.aws.amazon.com/s3/buckets/outputs-demo-e63a1b07"
bucket_info = {
  "arn" = "arn:aws:s3:::outputs-demo-e63a1b07"
  "name" = "outputs-demo-e63a1b07"
  "region" = "ap-south-1"
}
bucket_name = "outputs-demo-e63a1b07"
db_password = <sensitive>

✓ Apply complete! Resources: 3 added, 0 changed, 0 destroyed.
```

3 Ways to Read Outputs

● terraform output (all)

```
terraform output
```

Shows all outputs — sensitive still hidden

● terraform output -raw (for scripts)

```
terraform output -raw bucket_name
```

outputs-demo-e63a1b07

No quotes — use directly in shell: BUCKET=\$(terraform output -raw bucket_name)

● terraform output -json (reveal sensitive)

```
terraform output -json db_password
```

"super-secret-password-123"

ONLY way to see sensitive value

SENSITIVE OUTPUTS

Hidden **Everywhere**

✗ HIDDEN — TERRAFORM PLAN

```
+ db_password = (sensitive value)
```

Value never shown at plan stage

✗ HIDDEN — TERRAFORM APPLY

```
db_password = <sensitive>
```

Value never shown after apply

✓ REVEALED — TERRAFORM OUTPUT -JSON

```
terraform output -json db_password
```

"super-secret-password-123" ← only this works

⚠ Production tip

Never store real passwords in Terraform. Use AWS Secrets Manager instead. SSM SecureString is fine for non-critical config.

TAKEAWAYS

Key Learnings



Outputs are shown after every terraform apply automatically



sensitive = true hides value in plan, apply, and output commands



-raw flag removes quotes — essential for shell script usage



-json flag is the only way to reveal a sensitive output value



Object outputs group related values — name + arn + region together



SSM SecureString encrypts value with KMS automatically — no extra config



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